

Conditional probability

If A and B are two events, the **conditional probability** that A occurs, **given that B has already occurred**, is written $P(A|B)$.

$P(A|B)$ is read as “the probability of A given B”.

To find $P(A|B)$, the sample space is reduced to B only, since B has already occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

EXAMPLE

A fair die is rolled. Find the probability that the number rolled is a five, given that it is odd.

The sample space for this experiment is the set $S = \{1, 2, 3, 4, 5, 6\}$ consisting of six equally likely outcomes. Let F denote the event “a five is rolled” and let O denote the event “an odd number is rolled,” so that

$$F = \{5\} \quad \text{and} \quad O = \{1, 3, 5\}$$

$$P(F|O) = \frac{P(F \cap O)}{P(O)}$$

Since $F \cap O = \{5\} \cap \{1,3,5\} = \{5\}$, $P(F \cap O) = 1 / 6$.

Since $O = \{1,3,5\}$, $P(O) = 3 / 6$.

Thus

$$P(F|O) = \frac{P(F \cap O)}{P(O)} = \frac{1 / 6}{3 / 6} = \frac{1}{3}$$

EXAMPLE 2

suppose that our sample space S is the population of adults in a small town The data are given in Table 2.1. One of these individuals is to be selected at random. **What is the probability a male is selected given that he is employed.**

	Employed	Unemployed	Total
Male	460	40	500
Female	140	260	400
Total	600	300	900

M: a man is chosen,

E: the one chosen is employed

$$P(M|E) = \frac{P(M \cap E)}{P(E)}$$

$$P(E) = \frac{600}{900} = \frac{2}{3} \quad \text{and} \quad P(E \cap M) = \frac{460}{900} = \frac{23}{45}.$$

$$P(M|E) = \frac{23/45}{2/3} = \frac{23}{30},$$

EXAMPLE 3

The probability that a regularly scheduled flight departs on time is $P(D)=0.83$; the probability that it arrives on time is $P(A)=0.82$; and the probability that it departs and arrives on time is $P(D \cap A)=0.78$. Find the probability that a plane (a) arrives on time, given that it departed on time. (b) departed on time, given that it has arrived on time.

- (a) The probability that a plane arrives on time, given that it departed on time, is

$$P(A|D) = \frac{P(D \cap A)}{P(D)} = \frac{0.78}{0.83} = 0.94.$$

(b) The probability that a plane departed on time, given that it has arrived on time, is

$$P(D|A) = \frac{P(D \cap A)}{P(A)} = \frac{0.78}{0.82} = 0.95.$$



EXAMPLE 4

Consider an industrial process in the textile industry in which strips of a particular type of cloth are being produced. These strips can be defective in two ways, length and nature of texture. For the case of the latter, the process of identification is very complicated. It is known from historical information on the process that 10% of strips fail the length test, 5% fail the texture test, and only 0.8% fail both tests. If a strip is selected randomly from the process and a quick measurement identifies it as failing the length test, what is the probability that it is texture defective?

SOLUTIONS:-

Consider the events L: length defective, T: texture defective

Given that the strip is length defective, the probability that this strip is texture defective is given by

$$P(T|L) = \frac{P(T \cap L)}{P(L)} = \frac{0.008}{0.1} = 0.08.$$

Sampling with replacement

What is the probability of drawing a red marble, then drawing a blue marble from this jar?

The probability of drawing a **red** marble = $2/5$.

The probability of drawing a **blue** marble = $1/5$.

These are independent events as the second event is not affected by the first. The probability of **BOTH** of these events is found by the

Multiplication Rule

$$P(\text{red then blue}) = P(\text{red}) \cdot P(\text{blue}) = 2/5 \cdot 1/5 = 2/25$$



Sampling without replacement

What happens if you draw the first marble and do NOT put that marble back in the jar before drawing the second marble? If the marble is not "**replaced**", the probability of the second drawing changes, since there are less marbles in the jar.

The probability of drawing a red marble = $2/5$

Do **not** put marble back in jar. Do **not** put marble back in jar..

The probability of drawing a blue marble is now = $1/4$.

the events are **dependent**. Probabilities DO affect one another.

With Replacement: the events are **independent**. Probabilities do NOT affect one another.

Without Replacement: the events are **dependent**. Probabilities DO affect one another.